CoE-XII

Aakash
Medical | IIT-JEE | Foundations
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NCERT BOOSTER TEST SERIES

Test – 02

Topics covered in various subjects:

Physics : Laws of Motion,Work,energy and Power

Chemistry: Chemical Bonding and Molecular Structure, States of Matter, Thermodynamics

Botany : **Morphology of Flowering Plant :** Parts of Flower, The Fruit, The Seed - Structure of Dicot seed, Structure of Monocot seed, Semi- Technical Description of a Typical Flowering Plant, Description of Some Important Families - Fabaceae, Solanaceae, Liliaceae, Anatomy of Flowering Plant, Cell: The Unit of Life

Zoology : Body Fluids and Circulation, Excretory Products and their Elimination, Locomotion and Movement

Choose the correct answer

1. To keep a body in uniform motion
   (1) An external force must act
   (2) An external force may act
   (3) External force should not act
   (4) Both (1) and (3)
2. According to Galileo motion of an object moving on frictionless horizontal plane
   (1) Must have acceleration
   (2) Must have retardation
   (3) It should move with constant velocity
   (4) All of these
3. **Statement (A):** The law of inertia was inferred by Galileo from observation of motion of ball on a double inclined plane
   Statement (B): The law of inertia was inferred by kepler from observation of motion of ball on a double inclined plane
   (1) Only statement (A) is correct
   (2) Only statement (B) is correct
   (3) Both statement (A) and (B) are correct
   (4) Neither statement (A) nor statement (B) are correct
4. **Statement (A):** If the net external force on the body is zero, its acceleration is zero.
   Statement (B): If the net external force on the body is zero then body must move with constant velocity
   (1) Only statement (A) is correct
   (2) Only statement (B) is correct
   (3) Both statement (A) and (B) are correct
   (4) Neither statement (A) is correct nor statement (B) is correct
5. An astronaut accidentally gets separated out of his small spaceship accelerating in inter stellar space of constant rate 50 m/s^2. The acceleration of the astronaut at the instant after he is outside the spaceship will be
   (1) 50 m/s^2 (2) 100 m/s^2
   (3) zero (4) Insufficient date
6. In which of the following cases the net forces acting on a body is not zero
   (1) A drop of rain falling down with constant speed
   (2) A crock of mass 10g floating on the surface of water
   (3) A car moving with constant speed of 20 km/h on rough road
   (4) A pebble of mass 0.05 kg is thrown vertically upward

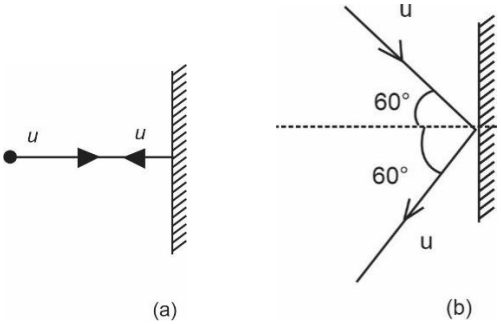
7. Suppose we are standing in a stationery bus and the driver starts the bus suddenly. We get thrown backward with a jerk is example of
   (1) Inertia of rest
   (2) Inertia of motion
   (3) Inertia of direction
   (4) Both 1 and 2

8. A large force acting for short time to produce a finite change in momentum is called
   (1) Impulse (2) Work
   (3) Acceleration (4) Power

9. A batsman hits back a ball straight in the direction of the bowler without changing its initial speed of 10 m / s. If the mass of the ball is 0.10 kg then the impulse imparted on the ball will be (assume linear motion of the ball)
   (1) 1 Ns (2) 4 Ns
   (3) 2 Ns (4) Zero

10. **Statement (A):** Force always occur in pairs. Force on a body A by B is equal and opposite to the force on the body B by A.
    Statement (B): There is no cause effect relation implied in the third law. The force on body B by A and force on body A by B act at the same instant
    (1) Only statement (A) is connect
    (2) Only statement (B) is correct
    (3) Both statement (A) and (B) are correct
    (4) Neither statement (A) nor statement (B) are correct

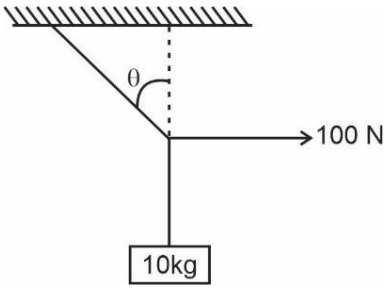
11. Two identical balls strike a rigid wall with the same speed but at the different angles, and get reflected without any change in speed as shown in figure (a) & (b). The ratio of magnitudes of impulse imparted to the balls by wall in situation (a) to the situation (b) will be



   (1) 1 (2) 1:2
   (3) 2 (4) 2:3

12. For translational equilibrium of a particle in mechanics refers to the situation
    (1) Net external force on the particle must be zero
    (2) Net external force on the particle must be non zero
    (3) Net external force on the particle may be zero
    (4) Net external force on the particle may be non-zero

13. A mass 10 kg is suspended by a rope of length 2m from the ceiling. A force 100 N in horizontal direction is applied at the mid point of rope as shown in figure. The angle subtended by rope with the vertical in equilibrium will be ($g = 10\ m/s^2$ and neglect the mass of rope)



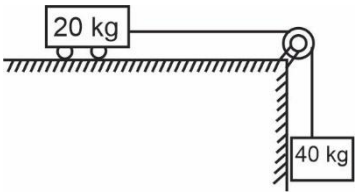
    (1) 30° (2) 60°
    (3) 53° (4) 45°

14. The magnitude and direction of the net force acting on a stone of mass 0.2 kg just after it is dropped from the window of the stationary train is ($g = 10 m/s^2$)
    (1) 2 N downward (2) 2N upward
    (3) 4N downward (4) 4N upward

15. The maximum acceleration of the train in which a box lying on its floor will remains stationary is (given that the co-efficient of static friction between the box and the train floor is 0.5 and $g = 10\ m/s^2$)
    (1) $2\ m/s^2$ (2) $5\ m/s^2$
    (3) $4\ m/s^2$ (4) $10\ m/s^2$

16. A mass of 10kg rest on a horizontal plane. The plane is gradually inclined until an angle $\theta = 30°$ with the horizontal. If the mass is just begins to slide. The coefficient of static friction between the block and the surface will be
    (1) $\frac{1}{\sqrt{3}}$ (2) $\sqrt{3}$
    (3) 0.5 (4) 0.2

17. A block of mass 40 kg and a trolley of mass 20kg is connected by a ideal string which is passing over an ideal pully as shown in the figure. If the system is released from rest then tension in the string will be (coefficient a friction between the trolly and surface is 0.05 and g = 10 m/s^2)



(1) 180 N (2) 280 N

(3) 140 N (4) 240 N

18. A car is moving on the banked rough road inclined at an angle θ with the speed $v_0 = \sqrt{Rg\tan\theta}$ (where R is the radius of curvature of the road). If speed of the car becomes $v < v_0$. Then friction force on the car will

(1) Act down to the slope

(2) Act up to the slope

(3) Not act

(4) Act down to the slope or up to the slope depends on the mass of the car

19. A circular racetrack of radius 300 m is banked at an angle of 45°. If the coefficient of friction between the wheels of the race-car and race track is 0.5. Then maximum permissible speed of the car to avoid the slipping will be

(1) 30 m/s (2) $30\sqrt{10}$ m/s

(3) 20 m/s (4) $20\sqrt{2}$ m/s

20. One end of the string of length *l* is connected to a particle of mass *m* and other to a small Peg on a smooth horizontal table. If particle move in a circle with speed *v.* Then tension in the string will be

(1) Zero (2) $\frac{mv^2}{l}$

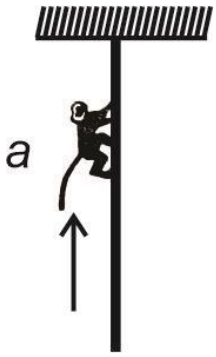
(3) $\frac{2mv^2}{l}$ (4) $\frac{mv^2}{2l}$

21. A bob of mass 0.2 kg hung from the celling of a room by string 2m long is set into oscillation. The speed of the bob at its mean position is 2 m/s. If the string is cut when the bob is at one of its extreme position. Then path of the bob will be

(1) Parabolic (2) Circular

(3) Straight line (4) Elliptical

22. A monkey of mass 30 kg climbs up on a light rope as shown in figure. If breaking strength of the rope is 400 N, then maximum acceleration of the monkey will be



(1) 5m/s^2 (2) $\frac{10}{3}$ m/s^2

(3) 20 m/s^2 (4) Zero

23. If $\vec{P} = 2\hat{i} + 3\hat{j} - 4\hat{k}$ and $\vec{Q} = 5\hat{i} + 2\hat{j} + 4\hat{k}$. Then angle between $\vec{P}$ *and* $\vec{Q}$ will be

(1) 30° (2) 60°

(3) 45° (4) 90°

24. If $\vec{A} = 2\hat{i} + 3\hat{j}$ and $\vec{B} = 2\hat{j} + 3\hat{k}$, then projection of $\vec{B}$ along $\vec{A}$ will be

(1) 6 (2) $\frac{6}{13}$

(3) $\frac{1}{6}$ (4) $\frac{6}{\sqrt{13}}$

25. Dot product of two vectors is a

(1) vector (2) tensor

(3) scalar (4) none of these

26. **Statement (A)**: scalar product obey a commutative law

Statement (B): scalar product obey distributive law

(1) Only statement is (A) correct

(2) Only statement (B) is correct

(3) Both statement A and B are correct

(4) Neither statement a statement are correct

27. The area under 'force displacement' curve gives

(1) Momentum (2) work

(3) Power (4) Impulse

28. A weightlifter holding 100 kg mass steadily on his shoulder for 10 seconds. The work done by the weight lifter during this time interval will be

(1) Positive

(2) Negative

(3) Zero

(4) May be positive or negative

29. A cyclist comes to a skidding stop in 2 m. During this process the force on the cycle due to the road is 100N and directly opposed the motion. The work done by the road on the cycle will be

(1) + 100 J (ii) – 100 J

(3) – 200 J (4) Zero

30. In a ballistics demonstration a police officer fire A bullet of mass 50.0 g with speed 200 m/s. on a soft plywood of thickness 2.0 cm. the bullet emerges with 50% of its kinetic energy. The emergent speed of the bullet will be

(1) 63.2 m/s (2) 141.4 m/s

(3) 100 m/s (4) 120 m/s

31. A body is falling freely under the action of gravity alone in vacuum. Which of the following quantity remains constant during the fall?

(1) Kinetic energy

(2) Protentional energy

(3) Total linear momentum

(4) Total mechanical energy

32. During head on elastic collision two bodies of mass m_1 and m_2 completely exchange their velocities when

(1) $m_1 << m_2$ (2) $m_1 >> m_2$

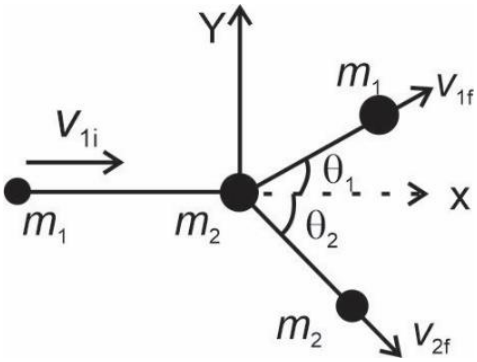
(3) $m_1 = m_2$ (4) $m_1 \leq m_2$

33. During head on elastic collision two bodies of mass m_1 and m_2. If just after the collision direction of velocity of m_1 reverse and magnitude of velocity remains same when

(1) $m_1 << m_2$ (2) $m_1 >> m_2$

(3) $m_1 = m_2$ (4) $m_1 \leq m_2$

34. Consider the collision as shown in the figure to be between to billiard balls with equal masses $m_1 = m_2$ to the first ball called the cue while second ball is called target. The billiard players wants to 'sink' the target ball in corner pocket, which is at angle $\theta_2 = 30°$ assume that the collision is elastic and that friction and rotational motion are not important then value of θ_1 will be



(1) 30° (2) 45°

(3) 60° (4) 90°

35. 1 kilowatt hour (kWh) of energy is equal to

(1) 3.6×10^6 J (2) 3.6×10^4 J

(3) 7.2×10^6 J (4) 7.2×10^4 J

36. Annual solar energy incident on the earth is

(1) 10^{68} J (2) 10^{44} J

(3) 3×10^{10}J (4) 5×10^{24} J

37. **Statement (A):** If total energy of the reactant is more than total energy of the product of the reaction. Then the reaction said to be exothermic

Statement (B): If total energy of the reactant is less than total energy of the product then reaction said to endothermic

(1) Only statement (A) is correct

(2) Only statement (B) is correct

(3) Both statement (A) and (B) are correct

(4) Neither statement A nor statement B are correct

38. The work done by all the forces (external and internal) on the system equals the change in

(1) Total energy

(2) Kinetic energy

(3) Potential energy

(4) None of these

39. Work done by a man in lifting a bucket out of a well by means of a rope tied to the bucket, will be

(1) Positive

(2) Negative

(3) Zero

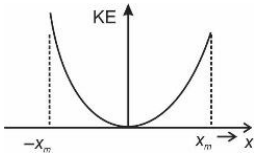
(4) May be positive or negative

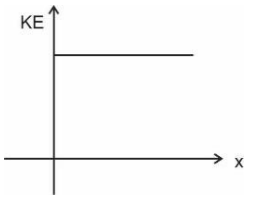
40. A body of mass 4 kg initially at rest moves under the action of an applied horizontal force of 10N on the table with coefficient of fraction (μ = 0.2). The net work done on the body in first 2 seconds will be (g = 10 m/s^2)
(1) + 2J (2) – 2J
(3) + 4J (4) + 6J

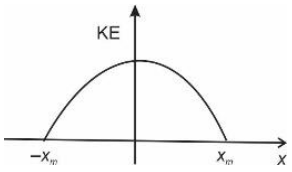
41. A body constrained to moved from point 'A(1,0,2) m to point B (2,3,5) m by action of a constant force $\vec{F} = (-\hat{i} + 2\hat{j} + 3\hat{k})N$. The work done by the force on the body will be
(1) 9 J (2) 15 J
(3) 14 J (4) 13 J

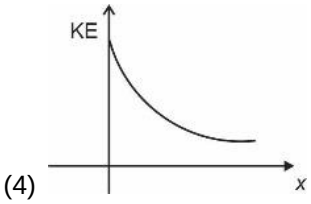
42. A body is moving unidirectionally under influence of a source of constant power. Its velocity in time *t* proportional to
(1) $t^{\frac{1}{2}}$ (2) t
(3) $t^{\frac{3}{2}}$ (4) t^2

43. Variation of kinetic energy of a block attached with an ideal spring best represented by which among the following graph?

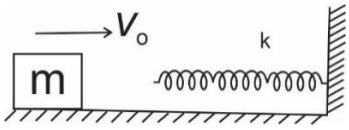
(1) 

(2) 

(3) 

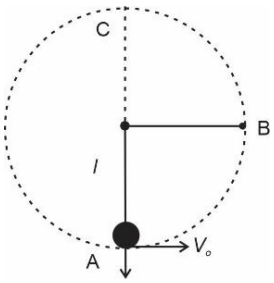
(4) 

44. A block of mass *m* moving with velocity V_0 on a smooth horizontal surface collide with an ideal spring having spring constant *k* attached with a wall as shown in figure. The maximum compression in the spring will be



(1) $V_0\sqrt{\frac{m}{2K}}$ (2) $V_0\sqrt{\frac{m}{K}}$
(3) $\frac{V_0}{2}\sqrt{\frac{m}{K}}$ (4) $V_0\sqrt{\frac{2m}{K}}$

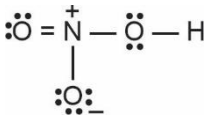
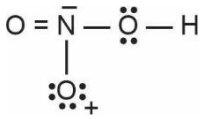
45. A bob of mass *m* is suspended by a light string of length *l*. It is imparted a horizontal *V*o at the lowest point A. Such that it just complete the vertical circle. The kinetic energy of the bob at position *B* as shown in the figure will be

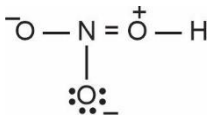
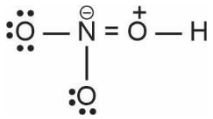


(1) $\frac{5}{2}$mgl (2) mgl
(3) $\frac{3}{2}$mgl (4) 2mgl

Chemistry

46. Correct lewis representation of HNO_3 is

(1)  (2) 

(3)  (4) 

47. Consider the following statements :
(a) Formal charges help in the selection of the lowest energy structure from a number of possible Lewis structures.
(b) The formal charge is factor based on a pure covalent view of bonding in which electron pairs are shared equally by neighbouring atoms.
(c) Lewis dot structures, represent the actual shapes of the molecules.

The incorrect statement(s) is/are :
(1) (a) & (b) only (2) (c) only
(3) (a), (b) & (c) (4) (b) & (c) only

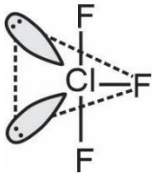
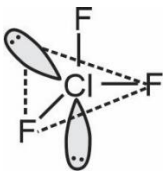
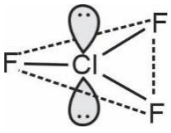
48. Decreasing order of bond length of given molecules, C = O, N = O, C = C, C = N is
    (1) C = C > N = O > C = O > C = N
    (2) C = C > C = O > N = O > C = N
    (3) C = N > C = C > N = O > C = O
    (4) C = C > C = N > N = O > C = O

49. Intramolecular H-bonding is present in
    (1) Phenol (2) p-nitrophenol
    (3) o-nitrophenol (4) m-nitrophenol

50. Which among the following statement is correct regarding resonance?
    (1) The canonical forms have real existence.
    (2) There is an equilibrium between the canonical forms.
    (3) Resonance averages the bond characteristics as a whole.
    (4) The energy of canonical structure is less than the energy of resonance hybrid.

51. For given molecule, the decreasing order of dipole moment is
    (1) $NH_3 > H_2O > H_2S > NF_3$
    (2) $NH_3 > NF_3 > H_2S > H_2O$
    (3) $H_2O > NF_3 > NH_3 > H_2S$
    (4) $H_2O > NH_3 > H_2S > NF_3$

52. Shape of SF_4 molecule is
    (1) Trigonal bipyramidal (2) T-Shape
    (3) See-saw (4) Square planar

53. Which arrangement of electrons is most stable?
    (1) 
    (2) 
    (3) 
    (4) All are equally stable

54. Which among the following features of hybridisation is correct?
    (1) The pure atomic orbitals are more effective in forming stable bonds than the hybrid orbitals.
    (2) The number of hybrid orbitals is not always equal to the number of the atomic orbitals that get hybridised
    (3) The hybridised orbitals are always equivalent in energy and shape
    (4) These hybrid orbital are non-directional in nature.

55. The unhybridised orbitals of carbon atom in C_2H_2. are
    (1) s and p_x (2) p_x and p_y
    (3) s and p_z (4) s and p_y

56. Hybridisation of Ni in $[Ni(CN)_4]^{2-}$ is
    (1) sp^3 (2) dsp^2
    (3) sp (4) sp^3d

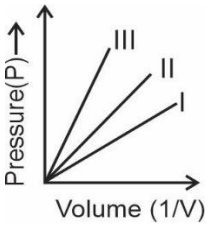
57. The electronic configuration of O_2 molecule is
    (1) $(\sigma 1s)^2(\sigma^*1s)^2\,(\sigma 2s)^2\,(\sigma^*2s)^2\,(\pi 2p_x^2 \equiv \pi 2p_y^2)$ $(\pi^* 2p_x^2 \equiv \pi^* 2p_y^2)$
    (2) $(\sigma 1s)^2(\sigma^*1s)^2(\sigma 2s)^2(\sigma^*2s)^2(\sigma 2p_z)^2\,(\pi 2p_x^2 \equiv \pi 2p_y^2)$ $\left(\pi^* 2p_x^1 \equiv \pi^* 2p_y^1\right)$
    (3) $(\sigma 1s)^2(\sigma^*1s)^2(\sigma 2s)^2(\sigma^*2s)^2(\sigma 2p_z)^2(\,\pi 2p_x^1 \equiv \pi 2p_y^1\,)$ $\left(\pi^* 2p_x^2 \equiv \pi^* 2p_y^2\right)$.
    (4) $(\sigma 1s)^2(\sigma^*1s)^2(\sigma 2s)^2(\sigma^*2s)^2\left(\pi 2p_X^2 \equiv \pi 2p_Y^2\right)$ $\left(\pi^* 2p_X^2 \equiv \pi^* 2p_Y^2\right)$

58. Bond order of C_2 molecule is
    (1) 0 (2) 1
    (3) 2 (4) 3

59. The hybrid orbitals used by carbon atoms in CH_3–CHO are
    (1) sp^2 and sp^2 (2) sp^3 and sp^3
    (3) sp^3 and sp^2 (4) sp and sp^2

60. The change in the hybridisation of B as result of the following reaction is
    $BF_3 + NH_3 \rightarrow F_3B.NH_3$
    (1) sp^2 to sp^3 (2) sp^2 to sp
    (3) sp to sp^3 (4) sp^3 to sp^2

61. I, II and III are three isotherms respectively at T_1, T_2 and T_3 as shown in the graph.



Correct order of temperature is
    (1) $T_3 > T_2 > T_1$ (2) $T_2 > T_3 > T_1$
    (3) $T_1 > T_2 > T_3$ (4) $T_3 > T_1 > T_2$

62. A balloon is filled with hydrogen at room temperature. It will burst if pressure exceeds 0.5 bar. If at 1 bar pressure the gas occupies 2.24 L volume, then at which volume can balloon be expanded?

(1) 4.48 L (2) 0.223 L

(3) 2.24 L (4) 1.12 L

63. A mixture of neon-dioxygen mixture contains 3.2 g dioxygen and 10 g Neon. If the total pressure of the mixture of the gases in the container is 50 bar, the partial pressure of O_2 is

(1) 10 bar (2) 8.33 bar

(3) 50 bar (4) 4.15 bar

64. The ratio among u_{mp}, u_{av}, u_{rms} is given by

(1) 1 : 1.128 : 1.224 (2) 1.224 : 1.28 : 1

(3) 1.224 : 1 : 1.28 (4) 1.28 : 1.224 : 1

65. van der Waals equation is

(1) $\left(P + \frac{an^2}{V^2}\right)(V - nb) = nRT$

(2) $\left(P + \frac{an^2}{V^2}\right)(V + nb) = nRT$

(3) $\left(P - \frac{an^2}{V^2}\right)(V - nb) = nRT$

(4) $\left(P - \frac{an^2}{V^2}\right)(V + nb) = nRT$

66. Gas is more difficult to compress when Z is

(1) 0 (2) $\frac{1}{2}$

(3) 1 (4) 2

67. Maximum critical temperature among the following is of

(1) H_2 (2) O_2

(3) CO_2 (4) NH_3

68. The temperature at which a gas obeys ideal gas law over an appreciable range of pressure is called

(1) Critical temperature

(2) Boyle temperature

(3) Inversion temperature

(4) Boiling temperature

69. The energy required to increase the surface area of the liquid by one unit is defined as

(1) Potential energy (2) Threshold energy

(3) Surface energy (4) Surface tension

70. The lowest imaginary temperature at which gases are supposed to occupy zero volume is

(1) Absolute zero (2) 273 K

(3) 273° C (4) 298° C

71. Which of the following statement is **incorrect** regarding assumptions of kinetic molecular theory?

(1) Collisions of gas molecules are inelastic

(2) Particles of a gas are always in constant random motion

(3) There is no force of attraction between the particles of a gas at ordinary temperature

(4) At any particular time, different particles in the gas have different kinetic energies

72. In van der Waals equation of gas, the constant 'a' is a measure of

(1) Magnitude of intermolecular attractive forces within the gas molecule

(2) Magnitude of intermolecular repulsive forces within the gas molecule

(3) Volume occupied by the molecules

(4) Speed of gaseous molecules

73. At 0°C, when the density of a certain oxide of a gas at 3 bar is same as that of dioxygen at 5 bar the molecular mass of the oxide is

(1) 53.3 (2) 45.5

(3) 19.2 (4) 20.7

74. The SI unit for the quantity PV^2T^2/n is

(1) $Nm^3K^2\,mol^{-1}$ (2) $Nm^3K^3\,mol^2$

(3) $Nm^4K^{-2}\,mol^{-1}$ (4) $Nm^4K^2\,mol^{-1}$

75. Volume occupied by 4.4 g of CO_2 at 0°C and 1 bar pressure is

(1) 2.2 L (2) 1.4 L

(3) 4.8 L (4) 7.2 L

76. Critical temperatures of four gases A, B C & D are 33.2 K, 154.3 K, 126 K and 405.5 K respectively. Which of these gases has stronger intermolecular forces?

(1) A (2) B

(3) C (4) D

77. Which of the following statements is/are correct?

(1) W_{ad} = +ve, when work is done on the system

(2) W_{ad} = –ve, when work is done by the system

(3) $\Delta U = W_{ad}$

(4) All of these

78. Heat is absorbed by the system at constant pressure is equal to

(1) ΔU (2) ΔH

(3) w (4) 0

79. In bomb calorimeter

(1) $\Delta T = 0$ (2) $\Delta V = 0$

(3) $\Delta P = 0$ (4) All of these

80 If $\Delta_{vap} H°$ for water at 298 K is 44 kJmol^{-1} then how much heat is required to evaporate 9g of water at 298 K?

(1) 11 kJ (2) 22 kJ

(3 99 kJ (4) 396 kJ

81. For which of the following molecule, standard molar enthalpies of formation at 298 K is zero

(1) Cl_2 (l) (2) CO_2 (g)

(3) Br_2 (l) (4) HCl (g)

82. Correct expression regarding a monoatomic gas M is/are

(1) $\Delta_r H^0$ (ionisation enthalpy) = E_o(ionisation energy) $+\frac{5}{2}RT$

(2) $\Delta_r H^0$ (electron gain enthalpy) = –A(electron affinity) $-\frac{5}{2}RT$

(3) $\Delta_r H^0$(ionization enthalpy) = $-E_o$(ionisation energy) + $\frac{5}{2}RT$

(4) Both (1) and (2)

83. Select the correct statement

(1) ΔS does not discriminate between reversible and irreversible process, whereas ΔU does

(2) ΔU does not discriminate between reversible and irreversible process, whereas ΔS does

(3) ΔU and ΔS both discriminate between reversible and irreversible process

(4) ΔU and ΔS both do not discriminate between reversible and irreversible process

84. The correct relation of $\Delta_r G°$ is

(1) $\Delta_r G° = +RT \ln K$

(2) $\Delta_r G° = + 2.303\ RT \log K$

(3) $\Delta_r G° = -RT \ln K$

(4) $\Delta_r G° = \Delta_r H° + T\Delta_r S°$

85. For the reaction,

$2P + Q \rightarrow R$

ΔH = 300 kJ mol^{-1} and ΔS = 0.5 kJ K^{-1} mol^{-1}. The temperature at which the reaction become spontaneous is

(1) Below 600 K (2) Above 60 K

(3) Above 600 K (4) Above 300 K

86. $\Delta U°$ of combustion of methane is –300 kJ mol^{-1}. The value of $\Delta H°$ is

(1) –300 kJ mol^{-1}

(2) Greater than –300 kJ mol^{-1}

(3) 0 kJ mol^{-1}

(4) Less than –300 kJmol^{-1}

87. Intensive property among the following is

(1) Internal energy (2) Heat capacity

(3) Volume (4) Temperature

88. For isothermal irreversible change work done is

(1) $-2.303\ nRT \log \frac{V_i}{V_f}$

(2) $-2.303\ nRT \ln \frac{V_f}{V_i}$

(3) $-P_{ex}\left(V_f - V_i\right)$

(4) 0

89 The entropy of any pure crystalline substance approaches zero as the temperature approaches

(1) 0°C

(2) 0 K

(3) 100°C

(4) 273°C

90. For one mole of NaCl(s)

$\Delta_{lattice} H°$ = +788 kJmol^{-1}

$\Delta_{hyd} H°$ = –784 kJmol^{-1}

The enthalpy of solution is

(1) –8 kJ mol^{-1} (2) –4 kJ mol^{-1}

(3) –10 kJ mol^{-1} (4) + 4 kJ mol^{-1}

Botany

91. Select the **incorrect** statement w.r.t. the outermost whorl of a typical flower.
(1) Is composed of petals
(2) Their members protect the flower in the bud stage
(3) Is calyx
(4) Their members are green and leaf like

92. The mode of arrangement of sepals or petals in floral bud w.r.t. other members of same whorl is known as
(1) Aestivation (2) Inflorescence
(3) Placentation (4) Phyllotaxy

93. Match the **Column I** with **Column II** and select the correct option.

	Column I	**Column II**
A	Staminode	(i) Petals united
B	Syncarpous	(ii) Sterile stamen
C	Gamopetalous	(iii) Sepals free
D	Polysepalous	(iv) Fused carpels

	A	B	C	D
(1)	(iv)	(iii)	(ii)	(i)
(2)	(iii)	(ii)	(iv)	(i)
(3)	(ii)	(iv)	(i)	(iii)
(4)	(i)	(ii)	(iii)	(iv)

94. In the flowers of which of the given plants, margins of petals overlap one another but not in particular direction?
(1) *Calotropis* (2) *Cassia*
(3) Pea (4) China rose

95. Which of the given is receptive surface for pollen grains?
(1) Anther (2) Stigma
(3) Style (4) Ovary

96. How many of the plants listed in box exhibit axile placentation in their ovary?

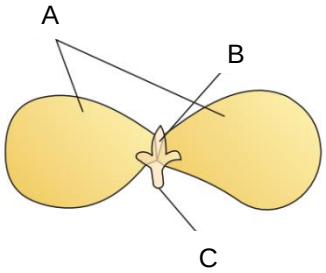
Pea, China rose, Tomato, Lemon, Primrose, *Dianthus*

(1) 2 (2) 3
(3) 4 (4) 5

97. In which of the given plants, the placenta develops at the base of ovary and a single ovule is attached to it?
(1) Marigold (2) Primrose
(3) *Argemone* (4) Mustard

98. Fruit is matured or ripened X, develops after fertilisation.
Select the option which correctly fills 'X'.
(1) Ovule (2) Ovary
(3) Radicle (4) Nucellus

99. Refer the given diagram and select the **correct** statement w.r.t. labelled parts A, B and C.



(1) 'B' is radicle
(2) 'C' represents plumule
(3) 'B' develops into root system on germination of seed
(4) 'A' is often fleshy and full of reserve food materials

100. Which of the given is medicinal plant of family Liliaceae?
(1) Ashwagandha (2) Belladonna
(3) *Aloe* (4) Muliathi

101. Prokaryotic cells differ from eukaryotic cells as the former
(1) Have cell wall
(2) Have cell membrane
(3) Contain ribosomes
(4) Lack nuclear membrane

102. Which one of the following cell organelles has its own DNA?
(1) Lysosome (2) ER
(3) Chloroplast (4) Lysosome

103. Which of the given statements is **incorrect** w.r.t. the organelles of endomembrane system?
(1) Is the major site of synthesis of lipid
(2) Produces cellular energy in the form of ATP
(3) Is the important site of formation of glycoproteins and glycolipids
(4) Is membrane bound vesicular structure formed by the process of packaging in Golgi apparatus

104. Live cells were first observed and described by
(1) M. Schleiden
(2) Robert Hooke
(3) Anton von Leeuwenhoek
(4) Robert Brown

105. Which of the given plastids contain chlorophyll and carotenoid pigments which are responsible for trapping most of the light energy, essential for photosynthesis?
(1) Chloroplast (2) Leucoplast
(3) Elaioplast (4) Aleuroplast

106. Which of the given plastids stores carbohydrate (starch) as in potato?
(1) Elaioplast (2) Amyloplast
(3) Aleuroplast (4) Chromoplast

107. Which of the given cell organelles is common to both prokaryotic as well as eukaryotic cells?
(1) Golgi apparatus (2) Ribosome
(3) Plastid (4) Mitochondrion

108. Which of the given chromosomes has a terminal centromere?
(1) Metacentric (2) Sub-metacentric
(3) Acrocentric (4) Telocentric

109. How many centriole usually a centrosome contains?
(1) 1 (2) 2
(3) 3 (4) 4

110. Golgi apparatus was first observed by
(1) Camilo Golgi (2) George Palade
(3) T. Schwann (4) Rudolf Virchow

111. The scar on the seed coat through which the developing seeds were attached to the fruit is called
(1) Hilum (2) Micropyle
(3) Testa (4) Tegmen

112. In maize seed, the outer covering of endosperm separates the embryo by a proteinaceous layer, called
(1) Scutellum (2) Aleurone layer
(3) Plumule (4) Radicle

113. % ♂ $K_{(5)}\ C_{1+2+(2)}\ A_{(9)+1}\ \underline{G}_1$ is the floral formula of which of the given plants?
(1) *Trifolium* (2) Mustard
(3) *Petunia* (4) *Asparagus*

114. In monocotyledonous seed, the plumule and radicle are enclosed in sheath which are called A and B respectively.
Select the correct option for A and B.

	A	B
(1)	Aleurone layer	Scutellum
(2)	Endosperm	Coleoptile
(3)	Coleoptile	Coleorrhiza
(4)	Fruit Wall	Seed Coat

115. In coconut, the fruit is known as drupe in which mesocarp is
(1) Fleshy (2) Stony
(3) Edible (4) Fibrous

116. Read the statements (A-C), stating them true (T) and false (F).
A. In eukaryotes, the ribosoms of chloroplasts are smaller than the cytoplasmic ribosomes.
B. Ribosomes are composed of RNA and proteins and are not surrounded by any membrane.
C. Cisternae are the infoldings in mitochondria.

	A.	B.	C.
(1)	T	T	F
(2)	T	F	F
(3)	F	T	T
(4)	T	T	T

117. Select the **incorrect** match w.r.t. cells and their different shapes
(1) Red blood cells – Round and biconcave
(2) White blood cells - Elongated
(3) Nerve cells – Branched and long
(4) Mesophyll cells – Round and oval

118. Which of the given group of bacteria are Comma-shaped?
(1) Bacillus (2) Coccus
(3) Vibrio (4) Spirillum

119. The A of a polysome translate the B into proteins.
Select the **correct** option for A and B.
(1) A – Ribosomes, B – RNA
(2) A – RNA, B - Ribosome
(3) A – ER, B - DNA
(4) A – DNA, B - ER

120. Ribosomes of mitochondria in animal cells are consist of
(1) 60 S and 40 S subunits
(2) 50 S and 30 S subunits
(3) 60 S and 30 S subunits
(4) 30 S and 40 S subunits

121. Which of the given tissues in grass helps to regenerate their parts removed by grazing herbivores?
(1) Lateral meristem (2) Intercalary meristem
(3) Secondary meristem (4) Apical meritstem

122. Select the **incorrect** statement w.r.t. Dicot leaf.
(1) Is dorsiventral
(2) Stomata is usually more in number on lower epidermis
(3) Mesophyll cell is differentiated into two types of tissues palisade and spongy parenchyma
(4) Vascular bundles are nearly similar in size due to presence of parallel venation

123. Which of the following is also referred to as 'starch sheath'?
(1) Pericycle of stem
(2) Epiblema of root
(3) Endodermis of stem
(4) Hypodermis of stem

124. Select the **incorrect** statement w.r.t. features of collenchyma.
(1) Occurs in layer below the epidermis in dicotyledonous plants.
(2) They do not provide any mechanical support to the plant parts
(3) Is found either as homogeneous layer or in patches
(4) Consist of cells which are much thickened at the corners due to deposition of cellulose, hemicellulose and pectin.

125. Conjoint closed vascular bundles are found in
(1) Dicot root (2) Monocot root
(3) Dicot stem (4) Monocot stem

126. Select the **incorrect** statement w.r.t. sclerenchyma.
(1) Consists of long, narrow cells with thick and lignified cell walls.
(2) Cells are usually living and contain protoplast
(3) Is simple tissue
(4) Found in the form of fibres or sclereids

127. Bast fibres
(1) Are also known as phloem fibres
(2) Are made up of collenchymatous cells
(3) Are absent in primary as well as secondary phloem
(4) Are multinucleated

128. Cork cambium is also known as
(1) Phelloderm (2) Phellem
(3) Phellogen (4) Pith

129. The tissue involved in secondary growth are
(1) Vascular cambium (2) Cork cambium
(3) Apical meristem (4) Both (1) and (2)

130. The tissues collectively known as periderm are:
A – Phellogen, B – Phellem, C – Phelloderm
The **correct** ones are:
(1) Only A and C (2) Only A and B
(3) Only B and C (4) All A, B and C

131. All tissues on the inner side of endodermis of dicot root such as pericycle, vascular bundles and pith constitute the
(1) Stele (2) Conjunctive tissue
(3) Casparian strips (4) Periderm

132. **Statement A**: Both apical and intercalary meristem are primary meristem.
Statement B: Secondary meristem occurs in mature region of roots and shoots of many plants.
Select the option which is true for the given statements.
(1) Only statement A is correct
(2) Only statement B is correct
(3) Both A and B are correct
(4) Both A and B are incorrect

133. The arrangement of ovules within the ovary is known as
(1) Inflorescence (2) Placentation
(3) Aestivation (4) Parthenogenesis

134. In plants, the middle lamella present between the adjacent cells mainly contains
(1) Cellulose (2) Hemicellulose
(3) Calcium Carbonate (4) Calcium Pectate

135. Select the **incorrect** statement w.r.t. heartwood.
(1) Comprises dead elements
(2) Has highly lignified cell walls
(3) Conducts water
(4) Gives mechanical support to stem

Zoology

136. Most abundant blood corpuscles are
(1) Neutrophils (2) Erythrocytes
(3) Platelets (4) Lymphocytes

137. Which of the following statement is **incorrect** w.r.t. circulatory fluids?
(1) Blood is a specialised connective tissue
(2) Sponges and coelenterates circulate water from their surroundings through their body cavities
(3) Lymph also helps in transport of certain substances
(4) Blood clotting factors are present in the plasma in an active from

138. Select the plasma proteins which are primarily involved in defense mechanisms.
(1) Albumins
(2) Globulins
(3) Fibrinogen
(4) Haemoglobin

139. Choose the **correct** match
(1) RBC – 6000–8000/mm^3 of blood
(2) Thrombocytes – 1,500,00-3,500,00/mm^3 of blood
(3) Haemoglobin – 12–16 mg/100 ml of blood
(4) WBC – 120 days life span

140. Which of the following statement is **incorrect** w.r.t. Rh grouping?
(1) Nearly 80% of humans are Rh positive
(2) Rh antibodies from the Rh positive mother can leak into the blood of the Rh negative foetus and destroy the foetal RBCs.
(3) Erythroblastosis foetalis can be avoided by administering anti-Rh antibodies to the mother immediately after the delivery
(4) Rh antigen is similar to one present in Rhesus monkeys

141. Given below is the table for the ABO grouping. Choose the option which **correctly** fill the blanks I, II, III and IV respectively

Blood group	**Antigens on RBCs**	**Antibodies in plasma**	**Donor's group**
A	**I**	**II**	A,O
O	**III**	**IV**	O

(1) Anti-A; B; A,B; anti A, B
(2) A; anti-A; nil; A, B
(3) Anti A; anti-B; nil; A, B
(4) A; anti B; nil; anti-A, B

142. Fibrins are formed by the conversion of inactive fibrinogens in the plasma by the enzyme
(1) Prothrombin (2) Thrombokinase
(3) Thrombin (4) Thrombocyte

143. Lymph can be best described as
(1) Plasma without clotting factors
(2) Blood without RBCs, WBCs and platelets
(3) Blood minus RBCs and proteins
(4) Lymph is blood without RBCs, larger plasma proteins and most of the formed elements.

144. Fats are absorbed in the
(1) Lacteals present in the intestinal villi
(2) Blood capillaries present in the intestinal villi
(3) Arteriole
(4) Venule

145. Choose the **mismatch** w.r.t. circulatory pathways
(1) Molluscs – Open circulatory system
(2) Annelids – Closed circulatory system
(3) All vertebrates – Possess a muscular Chambered heart
(4) Amphibians – Ventricle pumps Oxygenated blood only

146. Right atrio-ventricular opening is guarded by
(1) Bicuspid value (2) Mitral valve
(3) Tricuspid valve (4) Semilunar valve

147. Read the following statements and choose the option with **incorrect** statements only
(a) Nodal tissue has ability to generate action potentials without any external stimuli
(b) Nodal tissue is a specialised cardiac musculature
(c) The number of action potentials that could be generated in a minute is same at different parts of the nodal system
(d) The nodal tissue present in the upper right corner of the right atrium is also called pacemaker of the heart
(1) a,b and d (2) c only
(3) a and c only (4) b, c and d

148. Volume of blood pumped by each ventricle in a cardiac cycle is called

(1) Stroke volume

(2) Cardiac output

(3) End systolic volume

(4) End diastolic volume

149. Match Column-I with Column-II and choose the **correct** option

Column-I	**Column-II**
(a) Atherosclerosis	(i) Heart stops beating
(b) Cardiac arrest	(ii) Heart muscle is Suddenly damaged
(c) Heart failure	(iii) Ineffective pumping of blood
(d) Heart attack	(iv) Coronary artery disease

(1) a(ii),b(iv),c(i),d(iii)

(2) a(iv),b(ii),c(iii),d(i)

(3) a(iii),b(i),c(ii),d(iv)

(4) a(iv),b(i),c(iii),d(ii)

150. Which of the following is **not** the effect of sympathetic nervous system on heart?

(1) Increases heart rate

(2) Decreases speed of conduction of action potential

(3) Increases strength of ventricular contraction

(4) Cardiac output increases

151. Read the following statements and choose the **correct** option

(a) Ammonia diffuses across gill surfaces as ammonium ions

(b) Kidneys play a significant role in the excretion of ammonia

(c) Land snails are uricotelic

(d) Ammonia can be removed with minimum loss of water

	(a)	(b)	(c)	(d)
(1)	F	F	T	T
(2)	F	T	T	F
(3)	T	T	F	F
(4)	T	F	T	F

152. Choose the **mismatch** w.r.t. excretory structures

(1) Rotifers – Protonephridia

(2) Vertebrates – Kidneys

(3) Prawn – Antennal glands

(4) *Amphioxus* – Nephridia

153. Kidneys are situated between level of _____ and _____ vertebra, close to the dorsal inner wall of the ____ cavity. Choose the option which **correctly** fill the blanks respectively

(1) Third lumbar, last thoracic, pelvic

(2) Last thoracic, third lumbar, pelvic

(3) Last thoracic, third lumbar, abdominal

(4) Last lumbar, third thoracic, abdominal

154. How many nephrons are present in both the kidneys?

(1) 1 million (2) 2 million

(3) 300 million (4) 100 million

155. Which type of nephron is involved in urine concentration?

(1) Cortical nephron

(2) Nephrons in which vasa recta is absent or highly reduced

(3) Nephrons having short loop of henle

(4) Nephrons having long loop of henle

156. Choose the **odd** one out w.r.t. amount of blood filtered by the kidneys

(1) 1100-1200 ml of blood per minute

(2) $1/5^{th}$ of the blood pumped by each ventricle of the heart in a cardiac cycle

(3) 20% of the cardiac output

(4) 1/5 × (Stroke volume × heart rate)

157. Read the following statements and choose the **correct** option

Statement A : All constituents of plasma pass into the lumen of the Bowman's capsule.

Statement B : Glomerular capillary blood pressure causes filtration of blood through 3 layers.

(1) Both A and B statement are correct

(2) Only statement A is correct

(3) Only statement B is correct

(4) Both A and B statement are incorrect

158. The amount of filtrate formed by both the kidneys per minute is

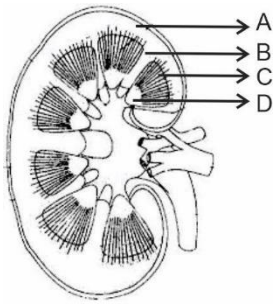
(1) 125 ml (2) 180 litre

(3) 1100–1200 ml (4) 18 litre

159. Which of the following is the part of JGA (juxta glomerular apparatus)
(1) All cells of DCT
(2) All cells of afferent arteriole
(3) Cellular modifications in DCT and afferent arteriole at the location of their contact
(4) Collecting duct

160. Select the **correct** statement for the part of renal tubule which is lined by simple cuboidal brush border epithelium
(1) Passive reabsorption of glucose and amino acids
(2) Active reabsorption of nitrogenous wastes like urea
(3) Minimum reabsorption
(4) 70–80% of electrolytes and water are reabsorbed.

161. Identify the labelled part where maximum osmolarity of interstitial fluid is present



(1) A (2) B
(3) C (4) D

162. Osmoreceptors are present in
(1) Hypothalamus
(2) Medulla oblongata
(3) Cerebrum
(4) Pons

163. ANF is released in response to
(1) Low blood pressure
(2) High blood pressure
(3) High osmolarity
(4) Low GFR

164. Read the following statements and choose the **correct** option

Statement A : Urine is released from the urinary bladder when voluntary signal is given by the CNS

Statement B : Micturition involves contraction of smooth muscles of the urinary bladder and simultaneous relaxation of the urethral sphincter

(1) Both A and B statements are correct
(2) Only statement A is correct
(3) Only statement B is correct
(4) Both A and B statements are incorrect

165. Choose the **incorrect** match w.r.t. elimination of excretory wastes
(1) Liver – Bilirubin, biliverdin, cholesterol, degraded steroid hormones, vitamins and drugs
(2) Saliva – Small amount of nitrogenous wastes
(3) Sweat – Sterols, hydrocarbons and waxes
(4) Lungs – Remove large amounts of CO_2

166. Locomotion and movement of food through cytopharynx in *paramecium* occurs with the help of
(1) Cilia
(2) Flagella
(3) Pseudopodia
(4) Streaming of protoplasm

167. Which type of muscle is involved in the transportation of gametes through the genital tract?
(1) Striated muscle
(2) Visceral muscle
(3) Voluntary muscle
(4) Muscle having branched muscle fibres

168. The portion of the myofibril between two successive 'Z' lines is considered as the functional unit of contraction and is called
(1) Sarcolemma
(2) Sarcoplasm
(3) Sarcomere
(4) Sarcoplasmic reticulum

169. Read the following statements and choose the option with **correct** statements only
(a) Anisotropic band contains only myosin filaments
(b) Actin and myosin filaments are arranged parallel to each other also to the longitudinal axis of the myofibrils
(c) In the centre of each 'I' band is an inelastic fibre called 'Z' line which bisects it
(d) Thick filaments are held together by a thin fibrous membrane called 'M' line
(1) a, b,c, d
(2) b,c and d only
(3) a and c only
(4) b and d only

170. A subunit of which protein masks the active binding sites for myosin on the actin filaments?
    (1) Troponin
    (2) Tropomyosin
    (3) 'F' actin
    (4) 'G' actin

171. Cross-bridges break when____binds to myosin head

    Choose the option which **correctly** fills the blank

    (1) ATP (2) ADP
    (3) Ca^2 (4) Mg^2

172. A motor neuron along with the muscle fibres connected to it constitute a
    (1) Neuromuscular junction
    (2) Motor-end plate
    (3) Motor unit
    (4) Synapse

173. Matrix of cartilage is slightly pliable due to
    (1) Calcium salts
    (2) Chondroitin salts
    (3) More blood supply
    (4) Hydroxyapatite

174. What is the total number of cranial and facial bones in an adult human?
    (1) 29 (2) 22
    (3) 80 (4) 126

175. A single U-shaped bone which is also included in the skull, is present
    (1) At the base of buccal cavity
    (2) At the base of nasal cavity
    (3) At the posterior part of skull
    (4) At the dorsal part of skull

176. Choose the **incorrect** statement w.r.t. human skeleton
    (1) First vertebra is atlas
    (2) All mammals have 7 cervical vertebrae
    (3) Ribs are bicephalic
    (4) 11th and 12th pair of ribs are floating ribs

177. Choose the **correct** match

(1) Clavicle	–	One curvature
(2) Scapula	–	At the level of 3rd to 7th rib
(3) Pectoral girdle	–	Acetabulum
(4) Collar bone	–	Articulates with acromion process

178. Cartilaginous joint is present in
    (1) Skull bones
    (2) Between vertebrae
    (3) Between carpals
    (4) Between carpal and metacarpal of thumb

179. Progressive degeneration of skeletal muscle mostly due to genetic disorder is
    (1) Myasthenia gravis
    (2) Muscular dystrophy
    (3) Tetany
    (4) Arthritis

180. Common cause for osteoporosis is
    (1) Low Ca^{+2} in body fluids
    (2) Decreased levels of estrogen
    (3) Accumulation of uric acid crystals
    (4) Auto-immunity

❑❑❑

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Test – 02

Answer Key

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3.	(1)	39.	(1)	75.	(1)	111.	(1)	147.	(2)
4.	(1)	40.	(1)	76.	(4)	112.	(2)	148.	(1)
5.	(3)	41.	(3)	77.	(4)	113.	(1)	149.	(4)
6.	(4)	42.	(1)	78.	(2)	114.	(3)	150.	(2)
7.	(1)	43.	(3)	79.	(2)	115.	(4)	151.	(4)
8.	(1)	44.	(2)	80	(2)	116.	(1)	152.	(4)
9.	(3)	45.	(3)	81.	(3)	117.	(2)	153.	(3)
10.	(3)	46.	(1)	82.	(4)	118.	(3)	154.	(2)
11.	(3)	47.	(2)	83.	(2)	119.	(1)	155.	(4)
12.	(1)	48.	(3)	84.	(3)	120.	(2)	156.	(2)
13.	(4)	49.	(3)	85.	(3)	121.	(2)	157.	(3)
14.	(1)	50.	(3)	86.	(4)	122.	(4)	158.	(1)
15.	(2)	51.	(4)	87.	(4)	123.	(3)	159.	(3)
16.	(1)	52.	(3)	88.	(3)	124.	(2)	160.	(4)
17.	(3)	53.	(1)	89	(2)	125.	(4)	161.	(4)
18.	(2)	54.	(3)	90.	(4)	126.	(2)	162.	(1)
19.	(2)	55.	(2)	91.	(1)	127.	(1)	163.	(2)
20.	(2)	56.	(2)	92.	(1)	128.	(3)	164.	(1)
21.	(3)	57.	(2)	93.	(3)	129.	(4)	165.	(3)
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23.	(4)	59.	(3)	95.	(2)	131.	(1)	167.	(2)
24.	(4)	60.	(1)	96.	(2)	132.	(3)	168.	(3)
25.	(3)	61.	(1)	97.	(1)	133.	(2)	169.	(4)
26.	(3)	62.	(1)	98.	(2)	134.	(4)	170.	(1)
27.	(2)	63.	(2)	99.	(4)	135.	(3)	171.	(1)
28.	(3)	64.	(1)	100.	(3)	136.	(2)	172.	(3)
29.	(3)	65.	(1)	101.	(4)	137.	(4)	173.	(2)
30.	(2)	66.	(4)	102.	(3)	138.	(2)	174.	(2)
31.	(4)	67.	(4)	103.	(2)	139.	(2)	175.	(1)
32.	(3)	68.	(2)	104.	(3)	140.	(2)	176.	(2)
33.	(1)	69.	(3)	105.	(1)	141.	(4)	177.	(4)
34.	(3)	70.	(1)	106.	(2)	142.	(3)	178.	(2)
35.	(1)	71.	(1)	107.	(2)	143.	(4)	179.	(2)
36.	(4)	72.	(1)	108.	(4)	144.	(1)	180.	(2)

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